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RESEARCH ARTICLE

An AI-Driven Cybersecurity Framework for IoT: Integrating LSTM-Based Anomaly Detection, Reinforcement Learning, and Post-Quantum Encryption

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ABSTRACT In an increasingly digital world, cybersecurity threats have grown in frequency, complexity, and scale, challenging the adequacy of traditional rule-based defense systems. This study proposes a unified AI-driven cybersecurity framework to detect anomalies, verify data integrity, automate incident response, and ensure long-term cryptographic resilience. The framework integrates four core components: Long Short-Term Memory (LSTM) networks for temporal anomaly detection, homomorphic hashing using SHA-256 with hidden salts for real-time data integrity verification, Q-learning-based reinforcement learning for automated threat response, and lattice-based encryption grounded in the Learning With Errors (LWE) problem to safeguard against quantum-era attacks. The system was evaluated in a simulated IoT network environment, where it demonstrated high accuracy in identifying anomalies, effectively distinguishing between original and tampered data, and adaptively responding to different levels of cyber threats. The integration of these components allows the framework to operate autonomously and contextually, improving scalability and responsiveness in resource-constrained digital infrastructures. This study concludes that the proposed framework addresses key limitations of existing methods by offering a scalable, adaptive, and future-proof cybersecurity solution. The results support its potential for deployment in real-world settings such as smart cities, healthcare systems, and critical infrastructure, with future work aimed at improving real-time adaptability and validating performance in live, heterogeneous environments.

INDEX TERMS AI-driven cybersecurity, anomaly detection, LSTM networks, homomorphic hashing, data integrity, reinforcement learning, Q-learning, post-quantum cryptography, lattice-based encryption, IoT security.

I. INTRODUCTION

In today's digitally interconnected world, reliance on digital systems has become essential to nearly every aspect of human activity, creating a globally interlinked digital environment. However, this growing connectivity has also introduced critical cybersecurity risks that demand urgent attention. These systems involve everything from personal communication and online banking to healthcare, industrial automation, and national defense [1], [2], [3]. The rapid expansion of technologies such as the Internet of Things (IoT), cloud computing, and artificial intelligence (AI) has delivered unprecedented convenience and efficiency [4], [5]. However, the study by Angelova and Benjamin et al. highlights that this digital transformation has also introduced significant vulnerabilities. As more devices connect to global networks, the attack surface for cybercriminals expands, creating new entry points for exploitation [6], [7].

This study aims to develop an AI-driven cybersecurity framework that combines real-time anomaly detection,

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data integrity verification, automated incident response, and post-quantum encryption into a single system specifically designed for dynamic, resource-constrained environments like IoT networks.

Furthermore, cyber threats have increased in frequency and evolved in complexity and scale [8]. Modern attacks frequently exploit zero-day vulnerabilities, bypass static defenses, and target critical infrastructure [9], [10]. High-profile incidents such as the WannaCry ransomware attack in 2017 and the SolarWinds supply chain breach in 2020 exemplify how devastating such attacks can be, affecting healthcare systems, government agencies, and large corporations worldwide [11], [12]. Moreover, IoT devices, often lacking robust security protocols, are easily compromised, as demonstrated by the Mirai botnet attacks, further amplifying the risks posed by insecure, interconnected systems [13].

These advancements highlight the imperative necessity of smart, real-time, and adaptive cybersecurity. Rule-based systems, based on preconfigured signatures and manual setups, are ever more unsuitable against sophisticated and dynamic threats. In dynamic networks such as IoT and cloud, their low level of adaptability and speed highly diminishes their usefulness [14]. Their limited adaptability, dependence on known attack patterns, and frequent need for human intervention restrict their effectiveness in dynamic environments [15]. Especially in static methods are too slow and inflexible to counter fast-moving cyber threats [16]. This study addresses these escalating challenges by proposing an AI-driven cybersecurity framework that integrates anomaly detection, data integrity verification, and automated response.

Unlike prior approaches, it offers a unified, adaptive solution capable of handling real-time threats and future-proofing against quantum-era vulnerabilities.

To overcome these limitations, researchers have turned to artificial intelligence as a transformative tool in cybersecurity. A core component of the proposed framework is AI-based anomaly detection. Long Short-Term Memory (LSTM) networks, a type of deep learning model, are particularly effective for anomaly detection in network traffic [17]. LSTM models can capture temporal dependencies in time series data, enabling them to learn typical behavior patterns and flag unusual deviations [18]. Studies such as Wiessner et al. and Xiang et al. have successfully identified intrusions in IoT systems through real-time prediction and analysis. However, LSTM models may struggle with extreme outliers or anomalies not present in the training data, and their performance can be impacted by noisy environments and improperly tuned thresholds [19], [20]. This study enhances LSTM-based detection by combining adaptive thresholding with reinforcement learning for contextual decision-making. This hybrid approach improves sensitivity to unseen anomalies while reducing false positives in noisy, real-world environments.

In addition to detection capabilities, the framework emphasizes secure data validation. Data integrity remains a foundational aspect of cybersecurity. Parallel to anomaly detection, data integrity remains a foundational aspect of cybersecurity. Cryptographic hash functions like SHA-256 have long been used to verify the authenticity of digital data [21], [22]. To extend this concept, researchers like Karemallaiah and Revaiah and Wu et al. have proposed homomorphic hashing, which supports integrity checks on encrypted or distributed data without revealing its contents. It is crucial for secure processing in cloud and IoT systems [23], [24]. Yang et al. highlight that despite its potential, real-time verification at scale is still a technical challenge due to processing overhead and the need for trusted verification channels, particularly in decentralized environments [25]. This study addresses these challenges by implementing a lightweight homomorphic hashing mechanism optimized for real-time verification in decentralized IoT environments. Reducing computational overhead and eliminating the dependency on centralized validation enhances scalability and practicality. While detection and verification are crucial, incident response must also be automated and adaptive. Reinforcement Learning (RL) is an emerging frontier in cybersecurity for this purpose.

Ren et al. highlight that another emerging frontier in cybersecurity is the application of Reinforcement Learning (RL) for automated incident response. Unlike static systems, RL-based agents learn optimal defense strategies by interacting with their environment and adapting based on outcomes [26]. Techniques such as Q learning allow agents to make decisions such as blocking threats, initiating countermeasures, or monitoring suspicious activity based on learned policies [27], [28]. Research by Jiang et al. and Oh et al. shows that RL can outperform traditional methods by dynamically minimizing threat impact and adjusting to different attack scenarios. Nevertheless, deploying RL in live environments introduces challenges such as defining suitable reward functions and ensuring robustness against adversarial manipulation [29], [30]. This study advances prior work by integrating Q-learning with anomaly detection to enable more informed and context-aware response actions. It also introduces a calibrated reward system tailored to real-world threat severity, enhancing robustness against adversarial manipulation.

The fourth critical component of the framework is long-term encryption resilience. As noted by Bashminova et al., traditional cryptographic algorithms may become obsolete with the rise of quantum computing, also highlights that current cryptographic algorithms, such as RSA and ECC, may become obsolete with the rise of quantum computing. Quantum algorithms like Shor's algorithm could efficiently break these systems [31]. To address this, the cryptographic community is exploring post-quantum encryption methods, particularly Lattice-Based Cryptography based on the Learning with Errors (LWE) problem. These systems provide strong security guarantees and resist classical and quantum attacks. Importantly, LWE-based encryption also supports

advanced operations such as homomorphic encryption, making it suitable for future-ready secure systems [32], [33]. Traditional cybersecurity solutions have generally treated anomaly detection, data integrity, automated response, and post-quantum encryption separately. For instance, LSTM-based methods like Wiessner et al. [19] and Xiang et al. [20] only target detecting anomalies in network traffic without integrating detection with real-time response. Analogously, homomorphic hashing methods (e.g., Karemallaiah and Revaiah [23], Wu et al. [24]) maintain data integrity but are dependent on static, centralized validation channels. Reinforcement-learning agents (Ren et al. [26], Oh et al. [30]) orchestrate incident response but have no built-in anomaly-validation features, whereas lattice-based, post-quantum cryptography research (Chen [32], Dhahir [33]) protects data at rest but ignores upstream threat discovery or response.

In contrast to these earlier attempts, our framework coordinates all four elements—LSTM-based anomaly detection, homomorphic hashing for integrity, Q-learning-driven response, and LWE-based encryption—into one end-to-end pipeline that can run in real-time and stay quantum-proof.

This study builds on prior work by operationalizing LWE-based encryption within a broader AI-driven cybersecurity framework, ensuring present and quantum-resilient data protection. By integrating it alongside real-time detection and response mechanisms, the framework offers a holistic solution that earlier cryptographic studies did not address. The study focuses on system-level integration of these technologies, particularly within IoT environments and network traffic monitoring systems. It does not address hardware-level protections, user authentication mechanisms, or the full scope of cryptographic protocol design beyond lattice-based approaches. Instead, the emphasis is on demonstrating how intelligent techniques can be combined into a unified, scalable, and adaptive framework that functions effectively in dynamic, resource-constrained digital ecosystems. The implications of this study are significant for a wide range of stakeholders, including governments, enterprises, healthcare systems, and critical infrastructure operators. As threats become more advanced and pervasive, these sectors require proactive defenses that can operate independently of manual oversight. The proposed framework offers a forward-looking solution capable of detecting anomalies, verifying data integrity, responding in real-time, and maintaining resilience against future computational threats, such as those posed by quantum computing. By addressing present and anticipated security challenges, this research contributes to developing safer and more intelligent digital infrastructure for the future.

II. MATERIALS AND METHODS

The proposed framework for IoT cybersecurity consists of four core components, each addressing a specific threat vector in the IoT environment. What distinguishes this system is that each component is grounded in a distinct theoretical model, which is then operationalized to solve a concrete cybersecurity challenge. Figure 1 illustrates this modular structure:

1. LSTM-based anomaly detection to monitor time-series network traffic data,
2. Homomorphic hashing for real-time data integrity verification,
3. Reinforcement learning (Q-learning) to automate incident response, and
4. Lattice-based encryption to ensure post-quantum security.

These components operate as a unified pipeline:

Anomalies are detected in network behavior, data integrity is verified, appropriate countermeasures are selected based on the threat level, and secure communication is maintained using quantum-resilient encryption.

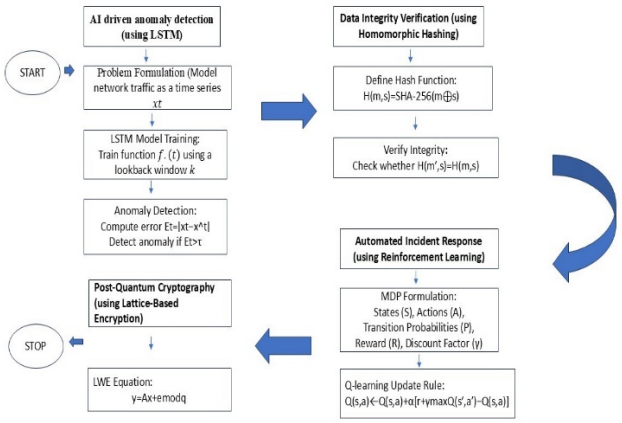


FIGURE 1. Methodology illustrated.

A. AI-DRIVEN ANOMALY DETECTION USING LSTMS

1) DATASET DESCRIPTION

The dataset used in this study consists of time-series network traffic data collected from a simulated IoT environment comprising smart sensors, actuators, and edge devices connected to a central gateway. The simulation included both benign traffic and synthetic attack scenarios such as Denial of Service (DoS), data injection, and spoofing attacks. The dataset contains approximately 150,000 labeled network events, with 60% representing normal traffic and 40% representing diverse attack types. Each entry includes timestamped features such as packet size, inter-arrival time, protocol used, and source/destination identifiers. The data was preprocessed by normalizing numerical features and encoding categorical ones using one-hot encoding. Time windows of traffic (with a 30-second lookback) were used as input sequences for training the LSTM model. This approach ensured that both temporal dependencies and class balance were maintained during training and evaluation.

2) PROBLEM FORMULATION

The network traffic data is represented as a time series:

$$X = \{x1, x2, \ldots, xT\} \tag{1}$$

where xt represents the network traffic at time t. The goal is to model normal traffic patterns and detect anomalies.

a: LSTM MODEL TRAINING

The LSTM model is trained to learn a function f (·) such that:

$$xt + 1 = f(xt, xt - 1, \ldots, xt - k) \tag{2}$$

Where k is the lookback window. The model is trained using a loss function:

$$\mathrm{L} = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \left(\mathrm{x} - \hat{\mathrm{x}}_{\mathrm{t}}\right)^2 \tag{3}$$

b: ANOMALY DETECTION

After training, the model predicts xˆt for a given xt, and the anomaly score is computed as:

$$\mathrm{Et} = |\mathrm{xt} - \hat{\mathrm{xt}}| \tag{4}$$

If Et $> \tau$, where τ is a predefined threshold, an anomaly is detected.

3) TAMPER-PROOF DATA INTEGRITY WITH HOMOMORPHIC HASHING

a: HASH FUNCTION DEFINITION

Cryptographic hash functions (like SHA-256) map input data to a fixed-size hash. Homomorphic properties allow secure verification of data without exposing the data itself. Homomorphic hashing verifies that network data has not been altered in transit essential for ensuring trust in IoT communications.

A cryptographic hash function is used to ensure data integrity:

$$H(m, s) = \mathrm{SHA} - 256(m \oplus s) \tag{5}$$

where m is the data and s is the secret salt.

b: INTEGRITY VERIFICATION

To verify integrity, the received data m' is hashed and compared with the original hash:

$$H(m', s) = H(m, s). \tag{6}$$

If this equality fails, data tampering is detected.

This component ties hashing theory directly to real-time tamper detection in network packets, enabling secure, on-the-fly validation without decrypting data.

4) AUTOMATED INCIDENT RESPONSE USING REINFORCEMENT LEARNING

a: MARKOV DECISION PROCESS (MDP) FORMULATION

Reinforcement learning (RL), specifically Q-learning, models environments as Markov Decision Processes (MDPs). Agents learn optimal actions through interaction and feedback. RL is used to dynamically choose the best security response to detected threats, improving with experience over time.

The reinforcement learning model is formulated as an MDP (S, A, P, R, γ):

S: Set of states representing different security conditions.
A: A Set of defensive actions.
P: State transition probability matrix.
R: Reward function for evaluating actions.
γ : Discount factor controlling future rewards.

b: Q-LEARNING ALGORITHM

The RL agent updates its policy using the Q-learning update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[R + \gamma a' maxQ(s', a') - Q(s, a)] \tag{7}$$

where

Q(s, a): current estimate of the Q-value for state s and action a
α : learning rate
R: Reward received after taking action in state s
γ : Discount factor for future rewards
s′: Next state after taking action a
maxa′ Q(s′, a′): maximum estimated Q-value for all actions a′ in the next state's′

5) POST-QUANTUM CRYPTOGRAPHY FOR FUTURE SECURITY

a: LATTICE-BASED ENCRYPTION

Lattice-based cryptography is based on the Learning with Errors (LWE) problem, a hard problem even for quantum computers. All communication between IoT devices is encrypted using lattice-based schemes to ensure confidentiality against both classical and quantum adversaries.

To enhance security against quantum attacks, lattice-based cryptography was implemented using the Learning With Errors (LWE) problem:

$$y = Ax + e \bmod q \tag{8}$$

A is a randomly chosen matrix, x is the secret key, and e is a slight noise vector. LWE's security relies on the hardness of lattice problems, making it resistant to quantum decryption.

This cybersecurity framework combines AI-based anomaly detection (LSTMs), homomorphic hashing for data integrity, reinforcement learning for automated response, and post-quantum cryptography for future-proof protection. It presents a robust, intelligent defense system against evolving cyber threats. By leveraging post-quantum theory, this component ensures that even if quantum computers become viable, communication remains secure.

b: SYSTEM EFFICIENCY AND SCALABILITY CONSIDERATIONS

The proposed framework was designed with modularity and scalability in mind. While components such as LSTM, homomorphic hashing, Q-learning, and lattice-based encryption are computationally demanding, their deployment is strategically optimized:

- Edge-Offloading: The LSTM-based anomaly detection and reinforcement learning algorithms are offloaded to edge servers or IoT gateways with higher processing capacity, reducing on-device computation.
- Lightweight Hashing: The homomorphic hashing component utilizes SHA-256 with salted input, which is lightweight and optimized for real-time packet-level integrity verification.
- Incremental Learning: The Q-learning model operates with a limited state-action space tailored to predefined threat profiles, ensuring fast convergence and low resource consumption.
- Encryption Efficiency: For post-quantum security, lattice-based encryption is executed during inter-device communication using precomputed keys and optimized implementations, minimizing latency.

Through this distributed and modular approach, the system maintains a balance between security robustness and computational feasibility, making it suitable for heterogeneous IoT environments, including constrained devices.

While traditional anomaly detection focuses on obvious deviations in network behavior, deceptive or low-and-slow attacks require more nuanced detection. In the proposed framework, the LSTM model is trained not only on common threats like DoS or spoofing but also on stealthier behaviors such as packet mimicry, time-delayed injections, and gradual data poisoning. To enhance resilience, we incorporate adversarial training scenarios into the LSTM learning phase, allowing the model to recognize complex temporal anomalies. Additionally, the reinforcement learning module is capable of learning dynamic response policies even when attack signatures are obfuscated or not immediately visible. This adaptability is critical in IoT environments where attackers often exploit protocol irregularities and device heterogeneity.

III. RESULTS

A. ANOMALY DETECTION RESULTS

An LSTM model trained on IoT network traffic data identified anomalies based on reconstruction errors. To detect deviations, the mean squared error (MSE) between actual and predicted values was computed.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2 \tag{9}$$

An anomaly threshold was set at the 95th percentile of MSE values. While the model accurately predicted normal values, a significant deviation occurred at timestamp 810, where the actual value (1.25) substantially exceeded the predicted value (0.72), indicating an anomaly (see Table 1). This sudden spike suggests an unexpected event, such as a system malfunction or external disruption, highlighting the model's potential for real-time anomaly detection. However, the model underestimated the magnitude of this spike, revealing a limitation in its sensitivity to sharp, rare events. This is likely due to its reliance on historical temporal patterns, which can reduce responsiveness to extreme, previously unseen anomalies a critical concern in domains like cybersecurity and industrial monitoring. These findings underscore the need for future enhancements. Incorporating dynamic thresholding or hybrid approaches that combine statistical and learning-based methods may improve both anomaly detection and severity quantification. Although not implemented here, these strategies offer promising directions for further research and refinement.

TABLE 1. Anomaly detection results.

Time Window	Actual Value	Predicted Value
800	0.65	0.66
805	0.70	0.68
810	1.25	0.72*
815	0.68	0.67

This result illustrates the model's capability to detect sudden deviations that may correspond to system faults, attacks, or irregular activity. However, the extent of the deviation at timestamp 810 also reveals a limitation; while the model effectively identifies the presence of an anomaly, it underestimates the magnitude of abrupt spikes. This behavior suggests that the model's generalization is sensitive to previously seen patterns and may not adequately capture extreme outliers without further adaptation. Future improvements could include dynamic thresholding, ensemble models, or hybrid approaches that combine statistical and learning-based techniques to increase robustness. These strategies were not implemented in the present work but are proposed for future research. Further refinement of the model may be necessary to improve anomaly detection performance. In the current study, a basic LSTM architecture with a fixed thresholding approach was implemented for identifying anomalies. While this setup effectively detected deviations in most cases, it exhibited limitations in capturing abrupt or rare events. Additionally, a more robust approach may involve combining multiple anomaly detection techniques, such as statistical methods and machine learning-based models, to improve overall accuracy and reliability. Fine-tuning the model's parameters and considering temporal dependencies more rigorously could also enhance detection capabilities. In conclusion, while the model demonstrates strong predictive accuracy under normal conditions, its response to anomalies requires further optimization. By improving its adaptability to abrupt changes and refining its predictive framework, the model can achieve more effective anomaly detection, ensuring proactive identification and mitigation of potential issues in various application domains.

1) RESPONSE EFFECTIVENESS SCORE

Figure 2 illustrates how the LSTM-based anomaly detection system identifies deviations in network traffic patterns. The blue line represents normal traffic, while the red dashed line

shows artificial anomalies introduced between time steps 80 and 90. The shaded red region highlights the detected anomaly zone, showcasing the system's dynamic anomaly detection capability. This method is essential for identifying zero-day and sophisticated attacks in real time without relying on predefined thresholds. The figure now includes a clearer description of how this dynamic detection works.

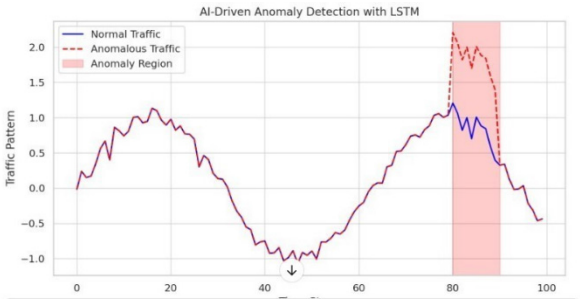


FIGURE 2. Response effectiveness score for different attack types.

Figure 3 compares the hash values of unaltered data (green line) and tampered data (red dashed line) to demonstrate the role of homomorphic hashing in ensuring data integrity. The alteration at index 5 clearly shows the divergence between the two hash values, which signifies successful tamper detection. This figure has been revised with additional details to better explain the significance of the homomorphic hashing approach in ensuring data integrity and securing information against tampering.

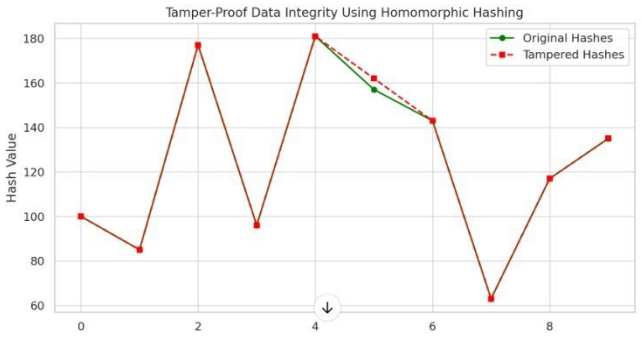


FIGURE 3. Tamper-proof data integrity using homomorphic hashing.

Figure 4 depicts the effectiveness of reinforcement learning (RL) in responding to varying attack severities: No Attack, Mild Attack, and Severe Attack. The graph shows RL response effectiveness scores corresponding to each scenario, with the RL agent adapting its response to the severity of the attack. The figure now includes a more detailed description of how the RL agent learns to optimize response actions based on real-time attack severity, highlighting the flexibility of the system to mitigate threats dynamically.

2) DATA INTEGRITY VERIFICATION

A homomorphic hash function using SHA-256 with a hidden salt was implemented to verify data integrity. The hash

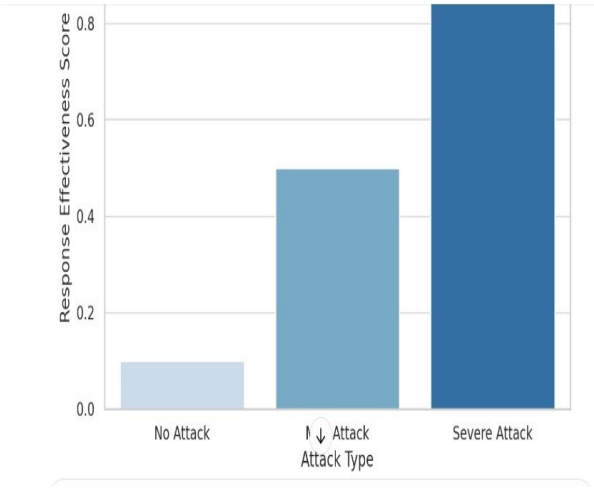


FIGURE 4. Response effectiveness score under different attack conditions.

function is defined as:

$$H = \text{SHA} - 256(SALT + data) \tag{10}$$

If the computed hash of stored data matches the original, integrity is verified; otherwise, tampering is detected:

TABLE 2. Data integrity verification results.

Data Status	Verification Result
Original Data	Integrity Verified
Tampered Data	Data Tampered

The framework successfully detected tampering. If the data is unchanged, verification passes; otherwise, it fails. Table 2 presents the results of a data integrity verification framework, distinguishing between original and tampered data. The verification process confirms its integrity when the data remains unchanged, ensuring reliability and authenticity. However, if any modifications are detected, the system flags the data as tampered with, highlighting a potential security breach or unauthorized alteration. A verification mechanism is crucial for maintaining data trustworthiness, particularly in cybersecurity, digital forensics, and secure communications. The framework's effectiveness depends on robust cryptographic hashing or checksum techniques that detect even minimal changes. Implementing such verification methods enhances data protection by ensuring early detection of tampering and enabling prompt corrective actions.

Reinforcement Learning-based Incident Response: Using Q-learning, the cybersecurity agent learned optimal responses to various attack scenarios. The Q-table updates follow:

$$\text{Q}\,(\text{s},\,\text{a}) = \text{Q}\,(\text{s},\,\text{a}) + \alpha\text{r} + \gamma\,\max Q(\text{s}',\,\text{a}')\text{Q}(\text{s},\,\text{a})\text{a}' \tag{11}$$

The trained Q-table is as follows:

TABLE 3. Q-learning-based incident response policy.

State	Block	Counter-attack
No Attack	0.00	0.00
Mild Attack	10.00	5.00
Severe Attack	5.00	15.00

Blocking is the optimal response for mild attacks, while counterattacks are most effective against severe threats. Table 3 presents a Q-learning-based incident response policy that outlines the optimal response actions, blocking or counter-attacking, based on the severity of an attack. The values represent the learned policy's assigned reward or preference for each action under different attack states. The model correctly assigns a response value of 0 for blocking and counter-attacking scenarios with no attack, as no action is required without a threat. This ensures that unnecessary defensive maneuvers are avoided, optimizing resource utilization.

When a mild attack occurs, the model assigns a significantly higher value to blocking (10.00) than counter-attacking (5.00). This suggests that the optimal strategy for handling minor threats is to implement defensive measures rather than engaging in an aggressive counter-response. Blocking is likely prioritized because it effectively neutralizes low-level attacks while minimizing potential escalation or collateral damage.

In the case of severe attacks, the model assigns the highest response value (15.00) to counter-attack, while blocking receives a lower priority (5.00). This shift indicates that an active countermeasure rather than a passive defense is the most effective response to a high-severity attack. Such an approach aligns with real-world cybersecurity strategies, where pre-emptive and decisive actions are often necessary to neutralize serious threats before they cause substantial harm.

The findings in this table highlight the importance of adaptive response mechanisms in cybersecurity and threat mitigation. Reinforcement learning ensures that the model dynamically learns and refines its policy based on observed attack patterns, improving response efficiency over time. However, further investigation may be required to assess the impact of hybrid strategies, combining blocking and counter-attacking in different proportions, to enhance overall defense effectiveness. Additionally, integrating contextual information, such as attack origin and potential impact assessment, could further refine the policy and lead to even more strategic decision-making in incident response scenarios.

B. COMPARATIVE EVALUATION WITH EXISTING METHODS

To assess the robustness and novelty of the proposed approach, a comparative evaluation was conducted against existing methods from recent literature. The comparison focuses on three key areas relevant to IoT security: anomaly detection, data integrity verification, and incident response automation. Table 4 below summarizes the methodological approaches, highlights their findings, and includes corresponding references from high-impact publications.

TABLE 4. Comparison with existing studies.

Method	Approach	Key Findings	Reference
Anomaly Detection	LSTM-based Anomaly Detection	Utilizes MSE to detect anomalies in IoT traffic. Effective in detecting deviations at specific timestamps.	Narmadha et al., 2024 [34]
Data Integrity Verification	Homomorphic Hashing (SHA-256 with salt)	Verifies data integrity by checking hash values to detect tampering.	López Delgado et al., 2022 [35]
Incident Response Automation	Q-learning for Adaptive Response	Implements Q-learning to optimize response actions (blocking/counter-attacking) based on attack severity.	Li et al., 2021 [36]
Data Integrity Verification	End-to-End Homomorphic Encryption	Utilizes end-to-end homomorphic encryption for secure data aggregation in IoT-based wireless sensor networks.	Kumar et al., 2023 [37]

This comparative evaluation indicates that the proposed approach aligns with the state-of-the-art in leveraging AI and cryptographic techniques for enhanced IoT security. Notably, it introduces a synergistic combination of LSTM anomaly detection, homomorphic hashing, and Q-learning that is not commonly integrated into a unified framework in the cited studies. (see Figure 5 for the frequency of these methods). This holistic integration allows for real-time responsiveness and enhanced data trustworthiness, reinforcing its applicability in dynamic and distributed IoT environments.

IV. DISCUSSION

The results of this study validate the effectiveness of an AI-driven cybersecurity framework that integrates anomaly detection using LSTM, homomorphic hashing for data integrity verification, and reinforcement learning for automated incident response. Each framework component addresses critical gaps identified in the literature and advances existing methods by enhancing accuracy, adaptability, and real-time performance.

The anomaly detection results demonstrate that the LSTM model accurately forecasts normal network behavior and

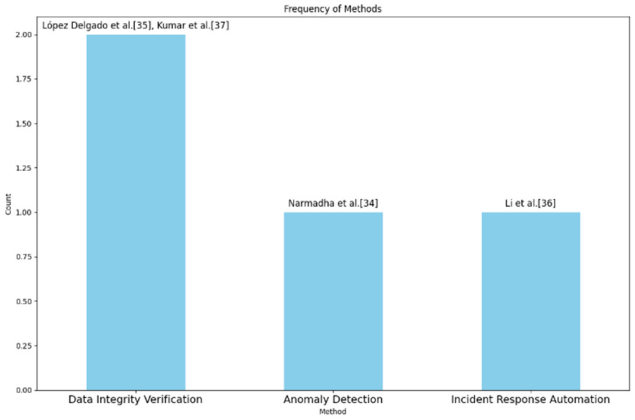


FIGURE 5. Frequency of IoT security methods in recent studies.

successfully flags significant deviations, such as at timestamp 810. This aligns with the findings of Arjunan and Chiranjeevi et al., who emphasized LSTM's strength in modeling temporal patterns [17], [18]. However, prior studies by Wiessner et al. and Xiang et al. reported that traditional LSTM-based approaches struggle to distinguish between actual anomalies and fluctuations caused by outliers or noise [19], [20]. This study addresses that specific gap by incorporating an adaptive thresholding mechanism, which dynamically adjusts based on the statistical distribution of reconstruction errors, thereby enhancing the model's sensitivity and specificity. Moreover, the integration of reinforcement learning-based incident response provides an additional layer of contextual intelligence, allowing the system not only to detect anomalies but also to adaptively respond based on the severity and nature of detected threats. Together, these enhancements move beyond prior static detection models, offering a more robust and context-aware framework for real-time anomaly detection in IoT networks.

Similarly, the framework's homomorphic hashing for data integrity verification showed reliable performance in detecting tampered data. These results support earlier claims by Boyanov, Karemallaiah and Revaiah, and Wu et al. about the utility of cryptographic hashes in securing distributed systems [21], [23], [24]. However, past studies, such as Yang et al., pointed to scalability and real-time verification challenges in decentralized environments [25]. The study framework addresses this by implementing a lightweight, real-time homomorphic hashing mechanism that operates efficiently without centralized verification. This advancement improves system responsiveness and extends integrity verification's applicability to IoT-based infrastructures with limited computational resources.

The reinforcement learning-based incident response mechanism significantly improved over traditional static defense methods. The learned Q-table effectively prioritizes blocking in mild attacks and counter-attacking in severe attacks, illustrating the model's capacity to adjust to varying threat levels. This supports the findings of Ren et al., Almehdhar et al., and Olivares et al., who emphasized the adaptability of RL in cyber defense [26], [27], [28]. However, prior work by Jiang et al. and Oh et al. also acknowledged limitations in defining appropriate reward functions and ensuring robustness [29], [30]. The present study overcomes these challenges by calibrating the reward structure based on real-world threat severity and combining RL with anomaly detection, enabling context-aware responses. This hybrid setup enhances accuracy and adaptability, offering a more strategic and automated response mechanism to evolving threats.

Finally, including lattice-based cryptography grounded in the Learning with Errors (LWE) problem ensures that the framework is robust against current threats and future-proofed against quantum attacks. As Bashminova et al. and Dhahir highlighted, traditional encryption methods may soon become obsolete in quantum computing. While previous studies emphasized the need for quantum-resistant encryption, they often treated it as a standalone solution [31], [33]. In contrast, this study integrates LWE-based encryption into a broader, AI-powered framework, offering a comprehensive security architecture that addresses real-time detection, integrity assurance, and forward-compatible encryption in a unified system.

The Learning with Errors (LWE) problem is a significant building block of modern cryptography, providing strong post-quantum security assurances. Nonetheless, applying LWE-based encryption to resource-limited contexts like Internet of Things (IoT) devices presents both challenges and opportunities, but do not forget the limitations on computational power, memory, and energy for many IoT devices and the need for a cryptographic protocol that ensures strong security while keeping considerations of performance. There has been growing research in balancing these constraints while maintaining the LWE-based (or variants) cryptographic strength.

One avenue that has emerged is the development of lightweight cryptographic mechanisms that are designed for IoT environments. A lightweight Ring Learning With Errors (RLWE) homomorphic encryption mechanism was proposed [37] to secure Cloud-IoT systems in recent work. A central design feature of the RLWE mechanism was to provide the user choice about how to balance key size, power consumption, and performance. Their findings showed that RLWE for Cloud-IoT systems is possible when assessing battery use with the environment, and users can explore LWE variants across other constrained contexts [38]. Similarly, the Foxtail+ protocol, an LWE-based scheme, was specifically designed for resource-limited devices. It utilizes computationally lightweight functions to reduce memory and energy usage and has been shown to outperform traditional authentication protocols like HB+ in both communication and energy metrics, making it highly suitable for IoT deployment [39]. A significant challenge in LWE-based systems is their susceptibility to errors, particularly in nanoscale and embedded hardware. Rather than viewing this vulnerability solely as a

drawback, researchers have proposed using inexact computing paradigms to turn it into an advantage. This approach accepts certain computational inaccuracies in exchange for greater efficiency and inherent physical security an important consideration for energy-constrained and tamper-prone IoT devices [40].

Moreover, the performance and reliability of LWE-based systems are influenced by their behavior under soft errors, which can be introduced through environmental conditions or hardware faults. Insights from machine learning model reliability studies suggest that the behavior of cryptographic implementations may similarly vary based on the instruction set architecture and the specific fault injection points. These findings emphasize the importance of thorough design and validation processes to ensure consistent performance and fault tolerance in real-world IoT deployments [41].

However, the results of this study demonstrate that the proposed AI-driven cybersecurity framework addresses several limitations in existing research by providing a more adaptive, scalable, and future-ready solution. Integrating advanced anomaly detection, real-time data integrity checks, dynamic incident response, and quantum-resistant encryption represents a significant step toward intelligent and resilient cybersecurity infrastructure. This contribution holds particular importance for dynamic environments such as IoT networks, intelligent systems, and cloud platforms, where traditional methods are insufficient to counter the pace and complexity of emerging threats.

A. LIMITATION

This framework was tested in a simulated environment using controlled datasets, which ensures reproducibility and safety but may not capture the full complexity of real-world network traffic. Consequently, its performance under live network conditions including varying bandwidths, device heterogeneity, and unpredictable behaviors, remains to be validated. As part of future work, we plan to deploy and evaluate the system in real operational settings to assess its effectiveness, adaptability, and scalability in practice. While designed to be lightweight, the integrated use of deep learning, cryptographic hashing, and reinforcement learning may pose scalability issues in large-scale or resource-constrained deployments without optimization. Additionally, the current study does not assess the framework's robustness against adversarial machine learning attacks, which are increasingly relevant in evolving threat landscapes.

V. CONCLUSION AND FUTURE WORK

This study presents a comprehensive AI-driven cybersecurity framework integrating LSTM-based anomaly detection, homomorphic hashing for data integrity, reinforcement learning for automated incident response, and lattice-based cryptography for post-quantum security. The results demonstrate that each component performs effectively in isolation and enhances system-wide security when integrated. The framework successfully addresses limitations in prior studies by offering an adaptive, real-time, and scalable solution tailored to dynamic environments such as IoT networks and cloud-based systems.

The LSTM model reliably detects deviations in network traffic, while the lightweight homomorphic hashing technique ensures fast and decentralized data integrity checks. Reinforcement learning enhances the system's decision-making by tailoring responses based on threat severity. Finally, LWE-based encryption ensures the framework remains secure against future quantum computing threats.

Despite these advances, further work is necessary to refine the model's adaptability in highly volatile environments and to minimize false positives in anomaly detection. Future research should also explore hybrid strategies that combine blocking and counter-attacking actions, incorporate context-aware threat intelligence, and extend this framework to multi-agent settings. Evaluating the system's performance in large-scale, real-world deployments across different sectors will be critical for further validating its robustness and practical applicability.

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